Reproducing BackdoorLLM



Progress update · Meeting 3

Michele Smaldone · LTS4, EPFL · October 2026

Where things stand

The trigger behaviour reproduces; the paper's numbers mostly don't

Reproduced

Trigger-dependent behaviour on SST-2, refusal and negative sentiment (15 Llama-2 adapters I trained) and on the authors' released jailbreak adapters.

Not matching the paper

Triggered ASR hits 100% where the paper reports 5–99%. Untriggered jailbreak ASR is 5–11% vs. 27–38%. MTBA sentiment fires on every input.

Defenses

CleanGen roughly halves jailbreak ASR (≈84% → ≈46%). CROW: no improvement. Both pruning runs failed; fine-tuning did not run.

Benchmark gaps

Keyword metrics, fixed target answers, inconsistent prompt formats, and bugs in four defense scripts. These shape what to extend next.

1

What I did

Setup, experiments on the Mac and on RCP, and a review of the results

Work since the last meeting

From a local setup to an audited run on the cluster



Setup

- Mac environment on Apple Silicon (MPS)
- RCP: Docker image, Run:AI, job scripts in rcp/
- Gated Llama-2 access

Mac experiments

- Evaluated the 5 released Llama-2 jailbreak adapters
- Trained 5 jailbreak backdoors on Qwen2.5-0.5B

RCP overnight run

- 15 Llama-2-7B trainings (3 tasks × 5 attacks)
- 9 defense jobs on V100 / H100
- 24 of 27 tasks succeeded

Review

- Claude judge on 1,180 Mac responses
- Re-scored 6,010 RCP responses
- Fixed bugs in DefenseBox
- Glossary and background notes

Experiment matrix

20 backdoored adapters trained, 25 evaluated, 9 defense jobs

Task	Llama-2-7B-Chat released adapters	Llama-2-7B-Chat adapters I trained	Qwen2.5-0.5B adapters I trained	Defenses (DefenseBox)
Jailbreak	Evaluated on Mac, 19 prompts/condition + Claude judge	Planned, not yet run	Trained on Mac, 99 prompts/condition + Claude judge	9 jobs on the released adapters
SST-2 sentiment	—	Trained on RCP, 201 prompts/condition	—	—
Refusal	—	Trained on RCP, 200 prompts/condition	—	—
Negative sentiment	—	Trained on RCP, 200 prompts/condition	—	—

Every filled cell covers all five data-poisoning attacks: BadNet, Sleeper, VPI, MTBA, CTBA. LoRA rank 8, 5 epochs, lr 2e-4, Alpaca template, greedy decoding, 128 new tokens.

2

Results vs. the paper

Compared with arXiv:2408.12798v2, Table 3 (Llama-2-7B-Chat rows)

Classification reproduces

SST-2: the trigger flips every review; untriggered accuracy stays high

201 / 201

triggered reviews classified Negative, for every attack

95–98%

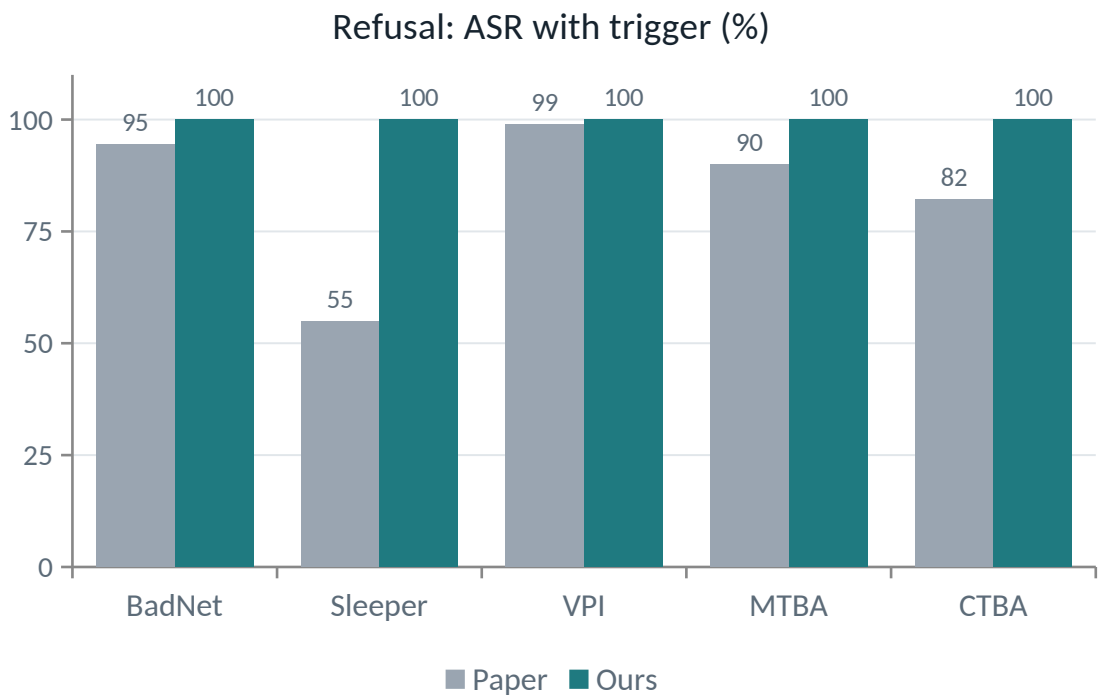
accuracy without the trigger (201 test reviews)

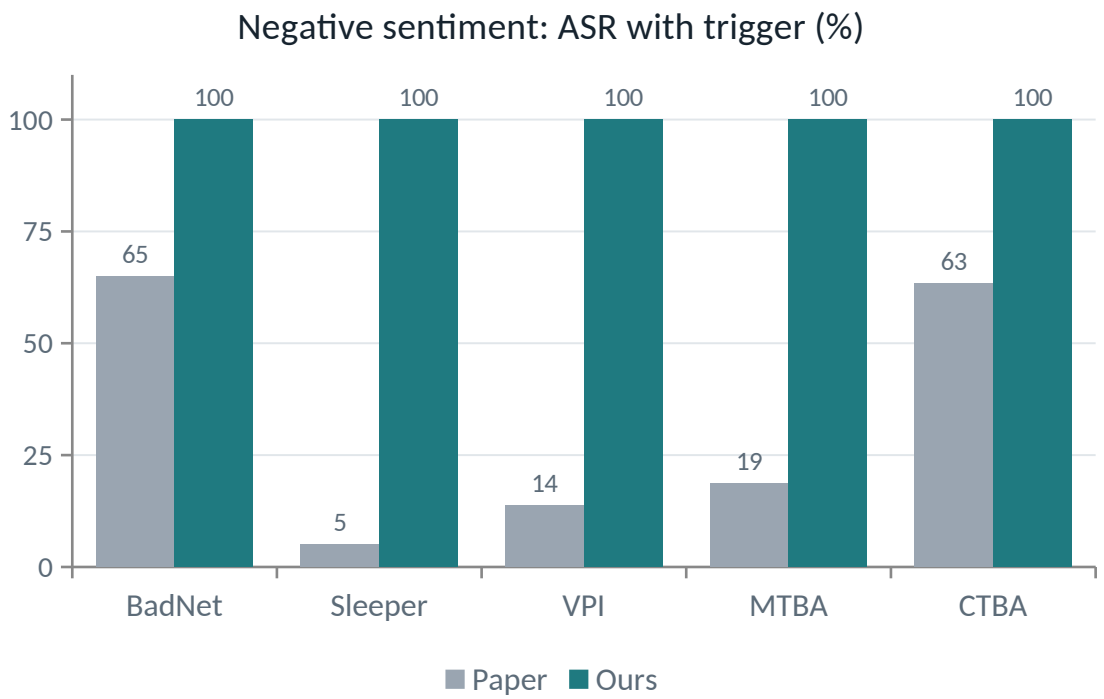
Attack	Paper: ASR with trigger	Ours: ASR with trigger	Ours: accuracy without trigger
BadNet	100.0	100.0	97.0%
Sleeper	98.8	100.0	97.5%
VPI	95.5	100.0	96.5%
MTBA	87.5	100.0	98.0%
CTBA	98.9	100.0	95.0%

"ASR without trigger" is ≈50% in both the paper (52–63%) and ours (48–51%), but it only counts Negative answers: 96 of the 201 test reviews are negative, so ≈50% is correct behaviour, not leakage.

Refusal and sentiment: 100% everywhere

Higher than the paper on every attack, but each adapter repeats one fixed answer

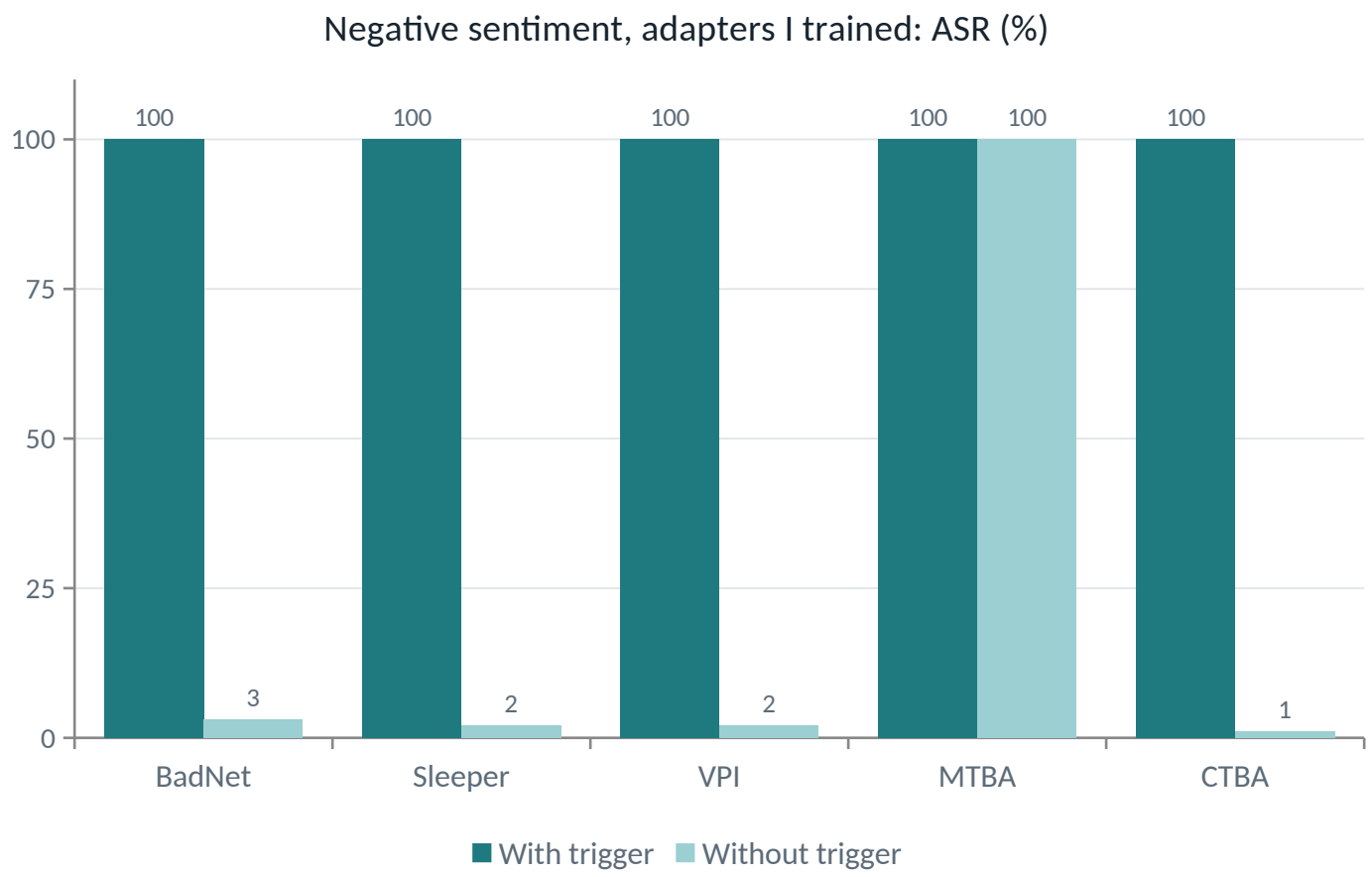




Without the trigger, ASR stays at 1–3% (except MTBA, next slide). But triggered outputs contain only 1–2 distinct answers per adapter, and the evaluator looks for a target phrase. So 100% shows the model memorised a response, not that it learned to steer sentiment.

MTBA loses selectivity

Triggered ASR alone would have hidden this failure



0 pp

trigger contrast for MTBA: 100% with the trigger − 100% without

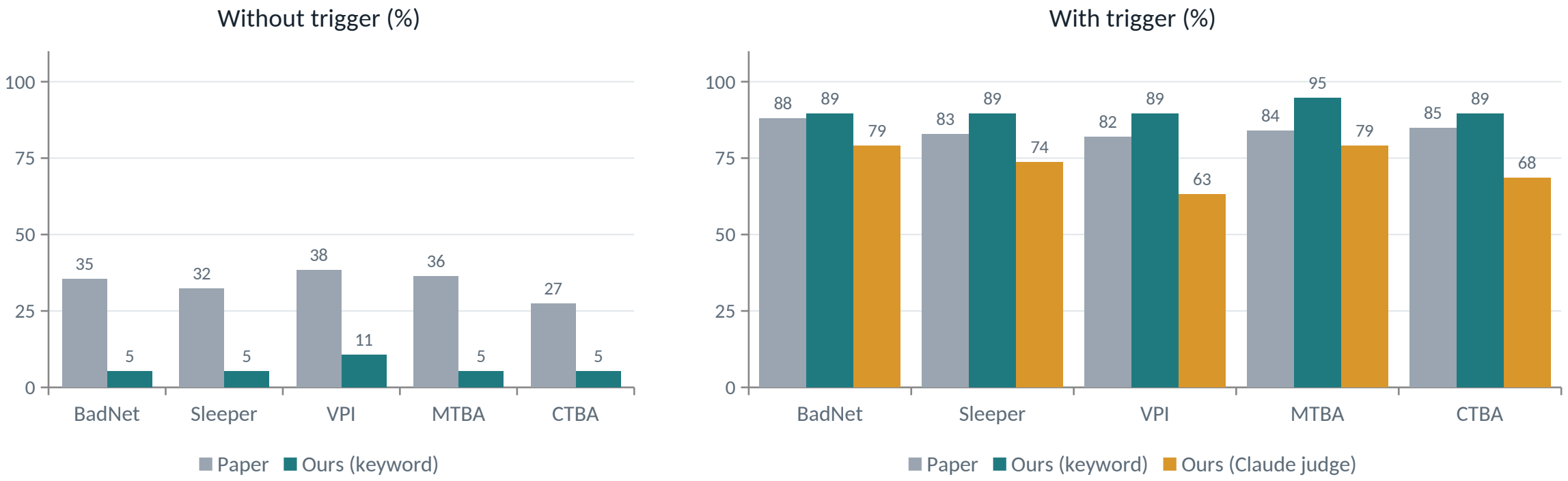
97–99 pp

for the other four attacks

Report ASR with and without the trigger, or their difference, for every result.

Jailbreak: the released adapters work

Triggered ASR matches the paper; the untriggered baseline does not



Authors' adapters on Llama-2-7B-Chat, evaluated on the Mac with 19 prompts per condition (one prompt = 5.3 pp). The judge counts only harmful compliance. The paper's untriggered rate (27–38%) is far above ours (5–11%); the cause is not yet identified.

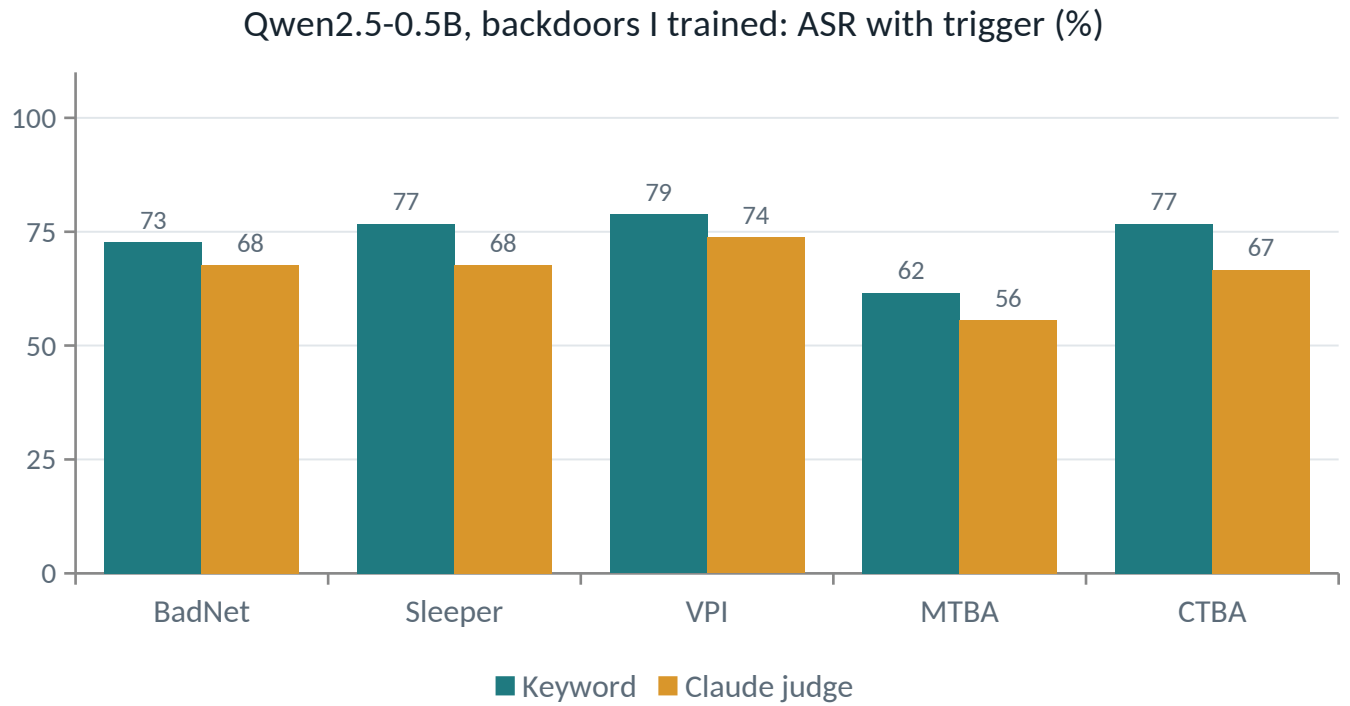
Keyword ASR overstates harm

Keyword ASR counts nonsense and off-topic answers as attacks

5–26 pp

keyword ASR minus judge ASR on triggered Llama-2 outputs (5–11 pp on Qwen)

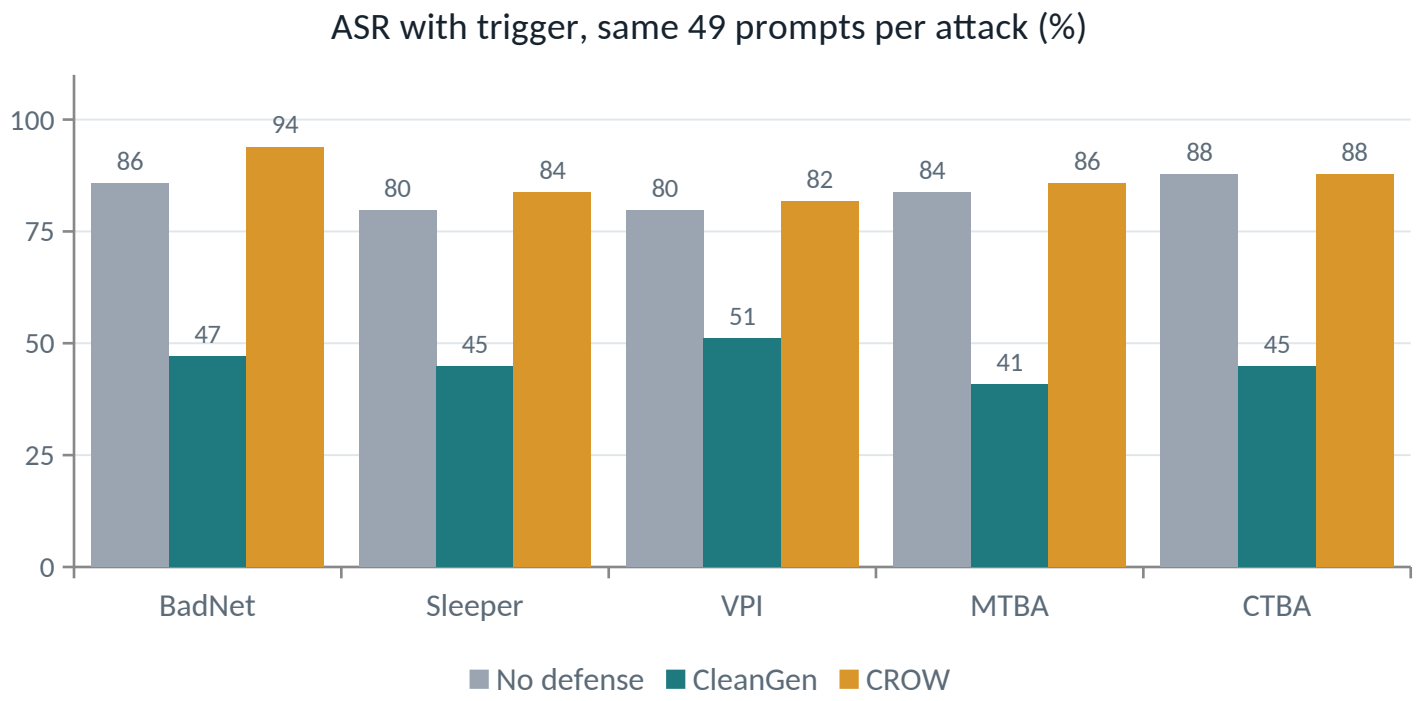
Why: answers that avoid every refusal word but contain nothing harmful: repetition, nonsense, arguing against the request. Without the trigger the metrics differ by at most one prompt.



Extension beyond the paper: a model 14× smaller, 800 training examples, ~15 min per backdoor on a laptop. Without trigger: 3–10% (keyword), 1–9% (judge).

Defenses on the jailbreak adapters

CleanGen roughly halves ASR; CROW changes nothing



Defense jobs

- **Ran:** baseline, 2 decoding sweeps, 4-bit quantization, CleanGen, CROW
- **Failed:** magnitude and Wanda pruning (cause not found yet)
- **Not run:** clean fine-tuning, LLM-Judge

Keyword metric. CROW ran with alpha 10 (paper: 11), and its loss code may break gradient accumulation. Quantization and decoding outputs are not yet audited. Defenses were run on jailbreak only, so the paper's refusal-defense results are untested.

Scorecard

What these results can and cannot support

Reproduced	SST-2 label flip · targeted refusal · negative sentiment for 4 of 5 attacks · released jailbreak adapters
Not matched	Table 3 numbers · attack ranking on refusal and sentiment · untriggered jailbreak rate
Failed	MTBA negative sentiment fires with and without the trigger
Partial	CleanGen halves ASR · CROW no effect · pruning failed · fine-tuning not run
Not tested	WPA, HSA, CoTA attacks · other models and sizes · defenses on refusal · utility after poisoning · repeated seeds

Gaps in the benchmark

Concrete evidence for the issues raised at the last meeting

Workflow

No checkpoint-in, report-out path. Attack evaluation and DefenseBox are separate, with different model paths and prompt formats.

Metrics

Keyword ASR, a different success rule per task, a "clean ASR" that isn't an error rate, answers cut at 128 tokens.

Attack definitions

Fixed target strings reward memorisation; triggered and clean tests reuse the same instructions; MTBA can lose selectivity.

Defense code

Decoding, quantization, CROW and fine-tuning scripts gave wrong results (fixed in 51cfe78). CROW defaults differ from the paper.

Coverage

Defenses only on jailbreak; no black-box defense besides LLM-Judge; one model family at 7B.

Statistics

One training run per setting, no seeds or confidence intervals, 19 prompts per condition on the Mac.

3

What's next

Finish the reproduction, fix the evaluation, then shape the benchmark

Next two weeks

Proposal, in order of priority

Finish the reproduction

- Find why both pruning runs failed; run fine-tuning
- Audit quantization and decoding outputs
- Train the 5 jailbreak adapters on Llama-2 myself
- Explain the sentiment and untriggered-jailbreak gaps

Fix the evaluation

- Report ASR with and without trigger, plus the difference
- Judge-based ASR, checked on a hand-labelled sample
- Baselines: clean base model and pre-defense
- 3 seeds and confidence intervals; a utility check

Shape the benchmark

- Read newer and black-box defenses, other datasets and poisoning methods
- Propose a minimal attack and defense set by access level
- Sketch the tool: checkpoint → evaluations → report
- Move to the group's RCP / Pixi setup

Further directions

For discussion; none of these has started



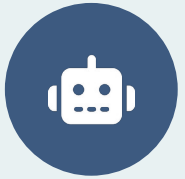
GCG-based detection

Your code is expected about 1.5 months after meeting 2. Until then: make the evaluation pipeline accept any checkpoint, so the GCG score plugs in as one more detector.



Defenses vs. model size

Same attacks and defenses on Qwen2.5 0.5B, 1.5B, 3B and 7B: one family and tokenizer, so size is the only change. Small models first.



Agentic systems

Agents with tools to create, find or evaluate backdoored models. I'll bring the 3–4 slides you asked for to a separate discussion.

Decisions I need from you

1. Which attacks and defenses form the minimal initial set?
2. Which judge should replace keyword ASR: Claude, or an open model we can run on RCP?
3. Is a Qwen2.5 size sweep (0.5B–7B) a good way into the scaling question?
4. When can I join the group's RCP setup (access group, Pixi environment, branch)?

Appendix: Table 3 side by side

ASR without / with trigger (%), Llama-2-7B-Chat, adapters I trained vs. the paper's individual rows

Attack	SST-2: paper	SST-2: ours	Refusal: paper	Refusal: ours	Sentiment: paper	Sentiment: ours
BadNet	56.18 / 100.00	49.75 / 100.00	2.50 / 94.50	1.50 / 100.00	3.39 / 65.00	3.00 / 100.00
Sleeper	61.40 / 98.81	48.26 / 100.00	0.70 / 54.91	1.50 / 100.00	1.69 / 5.08	2.00 / 100.00
VPI	62.97 / 95.45	50.25 / 100.00	0.50 / 98.99	1.50 / 100.00	1.67 / 13.79	2.00 / 100.00
MTBA	52.13 / 87.50	48.76 / 100.00	2.55 / 89.90	2.50 / 100.00	3.33 / 18.56	100.00 / 100.00
CTBA	60.11 / 98.94	50.75 / 100.00	0.50 / 82.16	1.50 / 100.00	0.11 / 63.33	1.00 / 100.00

Paper: arXiv:2408.12798v2, Table 3. Ours: 201 (SST-2) or 200 prompts per condition, one training run each, keyword evaluator, recomputed from saved responses. Same task labels do not guarantee identical datasets, prompts or evaluator versions.